

Poems and Songs. What can Textual Data Analysis do?

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Abstract

Textual statistical techniques have proven their effectiveness in areas such as the processing of responses to open questions in sample surveys, automatic analysis of messages on the Internet, topic modelling. Applications of these techniques to creative works, such as plays or novels (typologies, seriations and chronological series, authorship attributions) are also possible, owing to the usual large sizes of these texts. But the most difficult endeavor is perhaps the application to poetry, or, worse, to songs, which are generally short texts with fanciful or incongruous lexical frequencies (figures of speech, metaphors, onomatopoeia, and, mainly for songs, choruses). We will show, based on the corpus of 194 songs of the French poet-singer Georges Brassens (1921-1981), that the tools of the statistical analysis of textual data, suitably adapted, can contribute to describing and understanding a poetic work, despite its disconcerting features. Since these songs often have choruses leading to meaningless lexical frequencies, binary tables (presence-absence) should imperatively replace frequency tables. Analyses should be performed on both lemmatized texts and original texts. Correspondence analysis supplemented by Clustering (mainly Additive Trees in this example) remains a robust description tool, provided that Bootstrap techniques are used at each step to validate the results.

Keywords: Poetry, Textual Data Analysis, Correspondence Analysis, Clustering, Bootstrap.

1. Introduction

The exploratory analyses of textual data have shown their usefulness in several fields of application: a) In the case of short, numerous and qualified texts such as the sets of responses to open questions in sample survey together with responses to closed questions and respondent characteristics (meta-data); b) In the case of large texts, such as corpuses of novels, political speeches, chronological textual series (evolution, discontinuities, possible attributions of authors). In these two main cases, the lexical frequencies are meaningful. But most sets of poetic texts do not fit into these frameworks. In the case of collections of comparable poems, such as Shakespeare's 150 sonnets, it has been shown that several statistical tools can detect topics (Topic modeling) already identified by experts in the past (see for example: Lebart, 2018; Lebart *et al.*, 2019). But the corpus that we propose to study here is a more complex and elusive material: the collection of 194 songs written and sung by the French musician-poet Georges Brassens (1921-1981). This author is unique in that he brings together three almost contradictory features: a) He was a nonconformist and has rubbed shoulders with anarchist movements, b) In 1967, he received the poetry prize from the very conservative *Académie Française*. c) He was at the origin of the sale of 30 million records.

2. Problems and challenges of poetic texts

We will begin by showing how the statistical analysis of this type of text constitutes a methodological challenge. Then we will describe the pre-processing of the corpus, the statistical procedures used, and will discuss the first results obtained. We will keep in mind the

warning of Brunet (2004) during a remarkable statistical study of the poetic work of Arthur Rimbaud, a warning which applies also to the study of Brassens: *"Besides, the use of tools documentaries and statistics does not go without a certain naivety which gives its faith to the printed words, in their first innocence. But with Rimbaud, the words are often loaded. They are decoys, figureheads, and the reality they designate and hide eludes the most learned exegeses. When esotericism multiplies the traps, how to ensure the semantic constancy of the terms?"*. The studied works of Rimbaud included approximately 40,000 occurrences. The corpus of songs by Brassens used to exemplify the processing here has a comparable size: it contains approximately 52,000 occurrences. The poetic texts of Brassens are particularly rich in stylistic figures (litotes, metaphors, anaphors, euphemisms, allegories, etc.) which challenge the use of the word (being word-form or lemma) as a basic statistical unit. In that corpus, ribald, daring, even coarse slang words are sometimes inserted into the most subtle and tender poetry (Lamy, 2004, Poulanges and Tilleu, 2001). The poet brings up to date popular, outdated or slang expressions ("the poor man's coffee" for: sexual act, etc.). It often uses historical, obsolete terms (Rochard, 2009). In the case of songs that may include choruses or partial repetitions, versification constraints and rhymes, the lexical frequencies no longer have the statistical significance given to them in the lexical tables. Moreover, the texts are too short and the corpus too small to automatically extract new lexicographic units such as repeated segments, patterns, phrases.

The question we are asking may seem extremely narrow and technical in comparison with existing or potential works of literary and musical analyzes of the works of this musician-poet: "Do the typologies and visualizations obtained from the lexical profiles (approximately 1000 word-forms and as many lemmas) of the 194 songs, confronted with the available metadata, bring new information, relevant structural features or new materials likely to interest specialists?"

3. Preparation of texts and basic techniques

The 194 units of the corpus are described by as many groups of words (word-forms) and groups of lemmas which are complementary statistical units. These units are also described by the following metadata: a) Collection of albums (records) according to 14 categories, a variable which has a chronological component of publication (and often of composition/creation); b) The author according to 2 categories ("Brassens" [170 songs] or "Other", chosen and sung by Brassens [24 songs]); c) The dominant key (12 categories) of the musical score. The consideration of key is only a tentative experiment here. Note that the processing of musical scores through correspondence analysis has been studied, for example, in the pioneering works of Morando (1980), and Cocco (2014).

We will transform each song text, for both word-forms and lemmas, into an unweighted vocabulary, in other words, each element will only appear once within a given song (a few lines of Python easily allow this transformation of the text). The TreeTagger software (Schmid, 1994) has been used to obtain the lemmas. The lemmatized file has the advantage of reducing the diversity of inflections and therefore of allowing lower minimum frequency thresholds. It entails distances that are often more meaningful in terms of content whereas the file of word-forms retains the original diversity of forms, which is fundamental in the case of poetic texts. Thus, we will eventually obtain at each stage two different and complementary points of view.

Henceforth, the lexical tables (songs x words) will be tables of presence - absence. For this type of table, the correlation coefficient r between two songs coincides with the association

coefficient ϕ of Yule (1912). It is also linked to the χ^2 calculated on the contingency table (2 x 2) cross-tabulating the two texts described by n words, by the formula:

$$r^2 = \phi^2 = \frac{\chi^2}{n}$$

The techniques that can be used to describe and detect possible structures could be principal components analysis (licit on such binary coding), correspondence analysis (CA), clustering by additive tree (Huson and Briand, 2006), self-organized maps. The metadata defined above will act as active variables (grouping of songs according to records, for example), but above all as supplementary variables (*a posteriori* projection, with the indispensable Bootstrap validation). The software used, free of access, was DtmVic (www.dtmvic.com). Let us recall that in the context of binary coding, the eigenvalues do not have a straightforward interpretation in terms of information. Only the statistical validations (using the Bootstrap here) make it possible to judge the significance of the axes and the observed patterns.

4 Chronology and the sequence of records

The additive tree in figure 1 gives a summary of the links between the 140 records (discs, albums) identified by their year of publication. The last two years (1982_13 and 1985_14) are posthumous. These albums may include songs composed previously, or assortments suggested by the publisher, or even, early works found lately and interpreted by others. The first four years (from 1953_1 to 1958_4) constitute the lower right part of the tree.

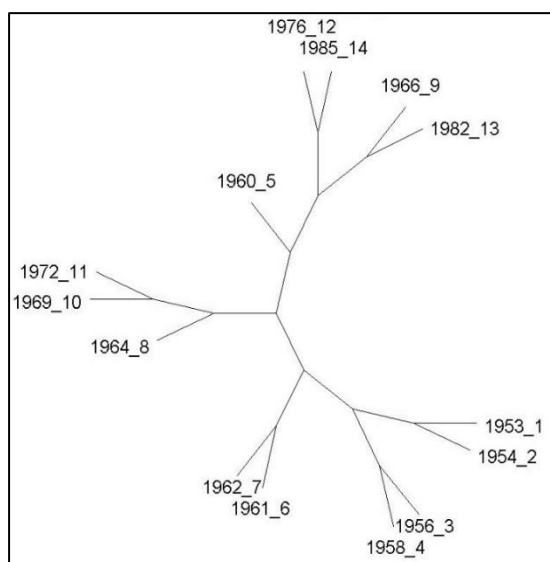


Figure 1. Additive tree of the 14 Brassens records (distances between records calculated from the presence-absence, within each song, of 943 word-forms).

There is a certain compatibility between the lexicographic similarities described by this display and the chronology. The opposition between the first years and the following ones will be found in all the analyses, whether they use lemmas or word-forms.

5 Brassens and his selected external poems

The 24 “external poems” set to music and sung by Brassens constitute a mini-anthology consistent with his personal values and his tastes. Figure 2 indeed shows the plan (1, 2) of the CA of the table crossing 194 songs and 901 lemmas (appearing at least four times).

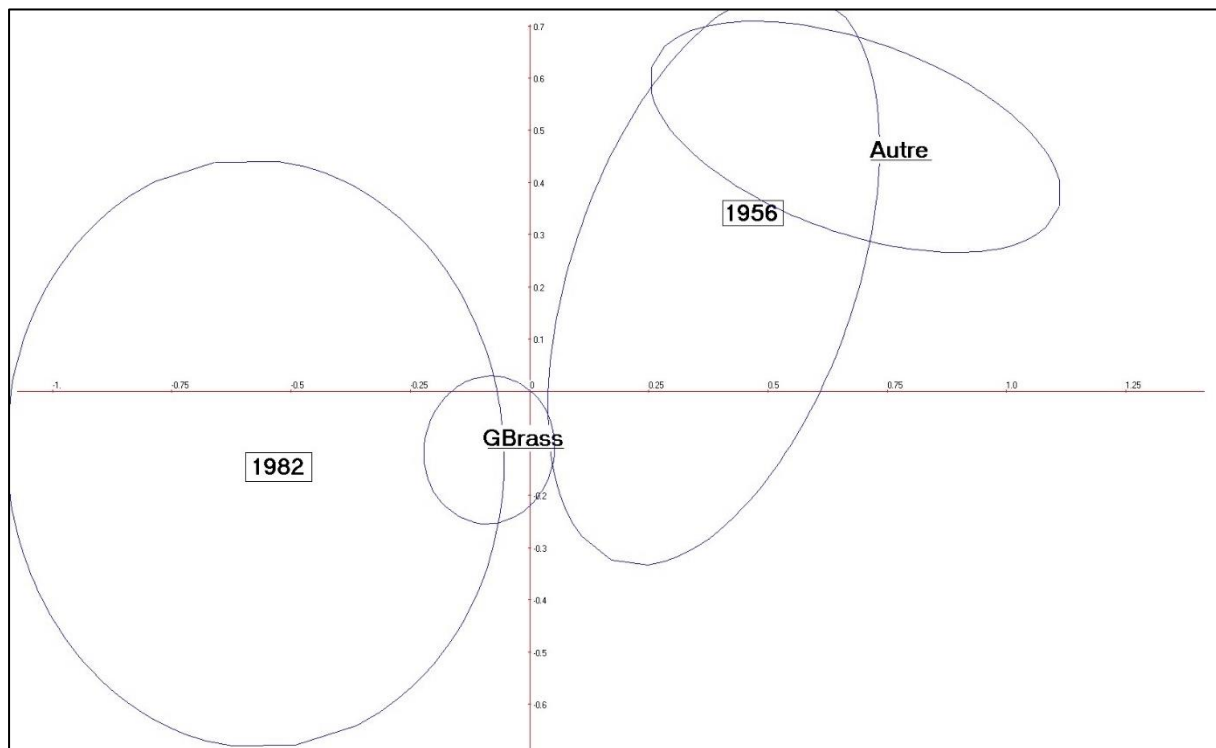


Figure 2. Principal plane from the CA (194 songs, 901 lemmas). Bootstrap confidence areas for four supplementary categories. **GBrass** (170 songs written by Brassens) and **Autre** (24 “external” poems set to music and sung by Brassens). Confidence areas for two records (**1982** and **1956**).

The two “author” points and the two years chosen are projected *a posteriori* in the plane of the CA (194 songs, 901 lemmas). The Bootstrap replications are obtained by drawings with replacement within the set of 194 songs. The “external” authors (Hugo, Lamartine, Paul Fort, Musset, etc.) are closer to the early years (1956). In fact, one could see on the factorial maps containing the locations of the 901 lemmas (unpublishable in this publication format, but which will be available for consultation in the www.dtmvic.com website) that the vocabulary has hardened over time, the author calling himself at the end a “pornographer” (from record 5). At the same epoch, the censorship, active for the first records, was more tolerant over time.

6. Keys

Brassens was a demanding composer-performer as well as a poet, and one could legitimately wonder if there was a link between the dominant keys and the characteristics of the songs represented by their lexical profiles. In the principal plane of the same CA (crossing 194 songs and 901 lemmas), the confidence areas of the 12 kept keys are indistinguishable except for G minor (Gm) on the left hand part of the display. However, this Gm exception only concerns five songs, three of which were posthumously performed by other singers. We will conclude, at this current state of research, that there are no significant links between the chosen key and the lexical profiles of the songs.

7. Confrontations between lemmas and word-forms

This phase is the most essential part when dealing with a poetic text. We will work on the 170 texts written by the singer himself. For most of the CA-type analyzes bearing on the 194 songs, or on the 170 songs of which Brassens is the sole author, a first dimension (horizontal on all the displays) dominates, whether it concerns lemmas or word-forms: It opposes ancient texts

(four to six first records) to more recent ones. Most of the ancient texts, like the external poems, have (often) a vocabulary that can be described as classic, even gallant, courtly or precious. The most recent have more cruder vocabulary. Figure 3, which concerns the 170 songs written and composed by Brassens, contains only a small excerpt of the set of the 872 active form words.

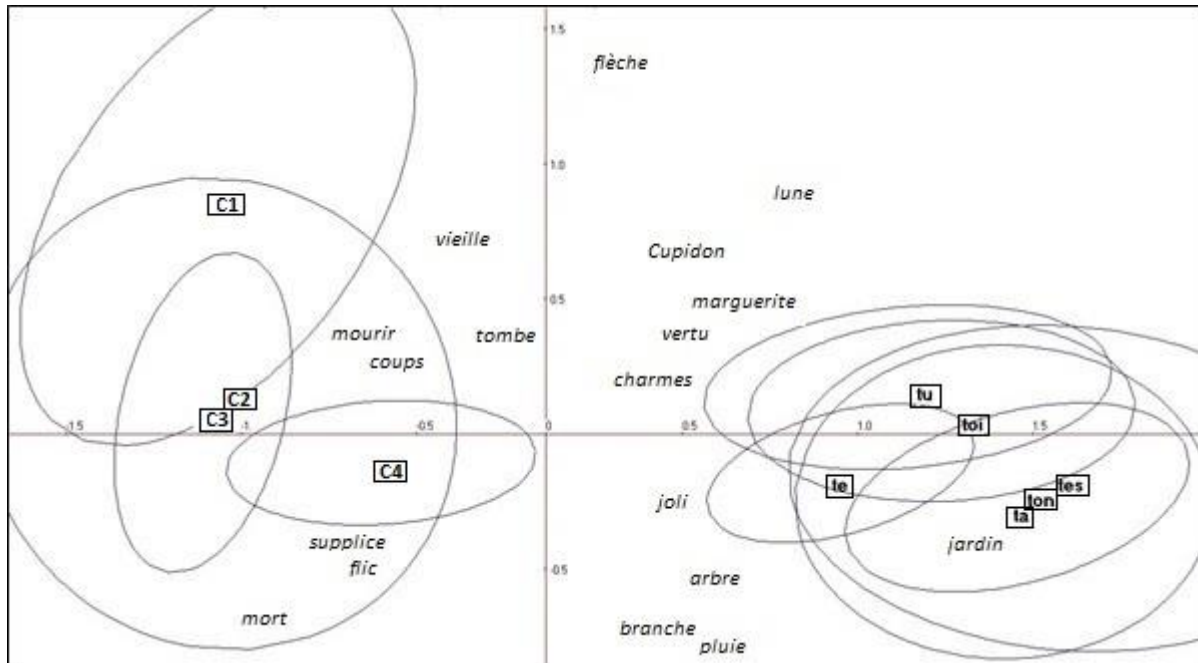


Figure 3. Word-forms (Brassens alone) (170 songs x 872 forms). In the foreground of the CA, opposition between the forms of familiarity for “you” in French (**te, tu, toi, ton, ta, tes**, on the right) linked to a refined poetry, and the presence of more bawdy vocabulary (on the left) **C1, C2, C3, C4** (cuckold, butts, ass, asses). Some Bootstrap confidence zones are drawn around the boxed item. The plan is also illustrated with some word-forms.

It shows that the familiarity (six forms of *you* concerned on the right) means above all intimacy, gentleness for Brassens (except for one song). Here, the word-forms reinforce the interpretation of this axis, which is similar to that obtained with the lemmas. In addition to the previous word-forms, we also find among the 20 rightmost points on this axis the words (arrow, daisy, moon, Cupid, virtue, charms, garden, rain, beautiful, tree, branch). On the left-hand part of the display, the four confidence areas concern words are less polite and are not commented here.

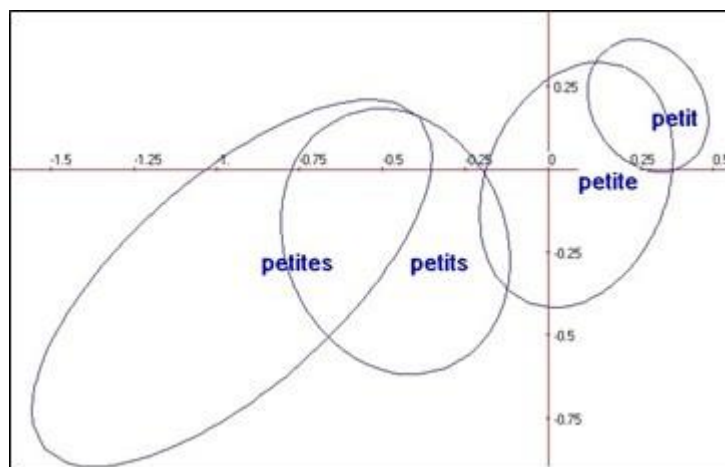


Figure 4. Bootstrap confidence zones of four inflections of the adjective “petit” (small) in the CA first plane of the presence-absence array (194 songs x 872 word-forms).

The word-forms play a role of magnifying glass for the interpretation, but they reveal also words that the lemmatization makes disappear, as shown in figure 4. In a CA performed on lemmas, the four inflections of the lemma "petit" in Figure 4 would be replaced by the unique lemma "petit", located in a central position which, in a way, neutralizes its contribution to the axis. These different inflections are nevertheless linked to the dimension which opposes, as we have seen, the intimate and the delicate (singular) to the impersonal and the bawdy (plural).

8. Conclusions

Despite the limited number of presented displays (necessarily small sized), one can guess that multivariate descriptive processing of poetic texts provides an original point of view on these texts and, above all, new material for analysis. Even from these first attempts of analysis, one could detect a general tendency, inextricably linked to age, career, personal evolution, perhaps to the increasing notoriety of the poet and probably to the growing permissiveness of the time. Each time, the use of forms amplifies, illustrates, and nuances the results obtained from the lemmas. The non-published graphs of the large additive trees representing the links between the 194 songs, the displays of principal planes from CA involving about 900 word-forms and as many lemmas together with the songs and the records, constitute also a set of working documents full of potential, difficult to publish on paper, but fascinating to consult for specialists or amateurs. We will risk parodying the adage "Garbage in, garbage out" (adage in fact very poorly adapted to data analysis techniques that do have a filtering function) by transforming it into "Poetry in, poetry out". Indeed, these new documents describing the intricate links and patterns between hundreds of words carefully chosen by the great troubadour Brassens are themselves a source of poetic emotions.

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